

Introduction

This project serves as a comprehensive exploration of digital literacy within the framework of modern engineering. In an era where "using" technology is common, this project focuses on becoming a "literate engineer"—someone who understands the underlying logic of systems, security, and professional communication. Over the course of five tasks, I transitioned from setting up a professional developer presence on GitHub to deploying an AI-driven application on Railway, conducting a peer-awareness quiz, and documenting critical cybersecurity case studies. This report details the technical implementation and the ethical considerations required to thrive in a global digital ecosystem.

Task 1 Write-up: The Development Phase

For this task, I developed "**walkthrough-ai**", a full-stack application designed to demonstrate the integration of backend logic with cloud hosting. The core of the project was built using **Python**, utilizing its robust data structure capabilities to handle user inputs and server-side requests. I used **Railway** as my primary deployment tool because it allows for seamless environment variable management and real-time logging.

Reflecting on this process, the most challenging part was not the coding itself, but managing the "invisible" layers of the application—specifically, ensuring that environment variables like API keys remained secure and that the server-side request handling was optimized. This task taught me that a successful project is defined not just by its UI, but by the stability and security of its backend architecture.

Task 2 Write-up: Professional Presence

I established a professional developer identity by creating a **special GitHub Profile README**. The repository, named after my username, acts as a digital business card. I integrated Markdown elements, including Unicode icons and structured headers, to clearly display my branch, year, and technical goals.

Moving forward, I plan to use this platform to host my academic assignments and personal projects like the **"Walk Through History"** itinerary generator. My future plans include automating this README using **GitHub Actions** to display my latest repository updates and technical certifications. By maintaining this profile, I am ensuring that my digital footprint reflects my growth as a Computer Science student and provides a centralized hub for my professional portfolio.

Task 3 Write-up: Platforms & Peer Engagement

For Task 3, I utilized **Google Forms and Google Sheets** to build a "Digital Literacy Awareness Quiz." I designed five targeted multiple-choice questions focusing on system logic, such as the purpose of .env files and the meaning of server error codes. I then linked this form to a Google Sheet to track real-time responses from my batchmates.

Practicing with these platforms highlighted the importance of data collection and organization. By analysing the response view in the linked sheet, I was able to identify common misconceptions among my peers regarding UPI safety and deployment protocols. This task bridged the gap between personal technical knowledge and community-level awareness, demonstrating how digital tools can be used for educational outreach.

Task 4 Write-up: Professional Communication

Communication in engineering requires a balance of clarity and etiquette. In this task, I drafted two distinct professional emails: an **extension requests** to a professor and an **internship inquiry** to a corporate coordinator.

The key learning here was the "3-Hour Rule" and the importance of a clear subject line. I learned that including specific technical details, such as my work with **Railway and Python**, makes an internship inquiry far more compelling. Professional communication is about respecting the recipient's time while providing all necessary context—like attaching a progress draft when asking for an extension. These templates now serve as my standard for all future academic and professional interactions.

Task 5 Write-up: Cybersecurity Awareness

What surprised me most during my research on **UPI/Online Payment Fraud** was how easily human psychology—rather than technical hacking—is exploited. I was startled to find that many victims willingly enter their UPI PINs because they simply do not know that a PIN is never required to *receive* money.

One habit I will change immediately is my approach to **public Wi-Fi**. I used to connect to open networks without a second thought; however, after learning about the risks of man-in-the-middle attacks and keyloggers, I have committed to using a VPN or my personal data hotspot for any sensitive activity. Becoming a literate engineer means being as protective of my data as I am of my code.

Conclusion

This project has been a transformative journey in defining my identity as a first-year engineer. Beyond the technical skills of Python coding and GitHub management, I have gained a deep appreciation for the ethical and professional responsibilities that come with digital literacy. I now view technology not as a set of tools to be used, but as a system to be understood and secured. The combination of backend development, professional branding, and cybersecurity awareness has provided me with a solid foundation for my remaining years in the Computer Science program.

References

- **GitHub:** github.com/Aklesh1206
- **Government Portal:** cybercrime.gov.in
- **Formatting Tools:** Shields.io, Google Forms, and Google Sheets.